Barcelona, Spain
albert.costa.sole@gmail.com · LinkedIn
ORCID 0000-0002-3126-0855 · Google Scholar · ResearchGate

Albert Costa Solé, PhD

Computational Engineer & Scientific Software · Multiphysics & HPC

PROFILE

Computational engineer with a PhD in numerical methods and 10+ years of experience in multiphysics modelling, high-performance numerical simulation and scientific software development. Technical background includes porous-media flow and reactive transport, electrochemical systems and biomedical modelling, with experience developing and running large-scale simulations on HPC systems. More recently, completed an MSc in Artificial Intelligence focused on machine learning, data-driven modelling and their integration with physics-based simulation.

TECHNICAL SKILLS

Programming	Python, Fortran, MATLAB
Scientific computing	Numerical methods, finite elements, multiphysics, porous-media flow, reactive transport
AI & machine learning	Supervised and unsupervised learning, deep learning, model selection and validation, data-driven modelling; scikit-learn, pandas, PyTorch (academic training)
HPC & software engineering	Linux, Git, CI/CD, TDD, MPI, Slurm; large-scale simulation workflows on HPC systems
Meshing & post-processing	GiD, gmsh, ANSA, ParaView
Simulation software	PFLOTRAN (contributor), PETSc-based solvers, Alya

PROFESSIONAL EXPERIENCE

Aug 2024 – Present

Senior Consultant, Amphos21, Barcelona

- **Design and run production-scale multiphysics simulations** for nuclear waste repository performance assessment, coupling flow, geochemistry and reactive transport on LUMI and JURECA and integrating site-specific hydrogeological and geochemical processes.
- **Develop and optimize scientific software** in Python and Fortran, with contributions merged into PFLOTRAN's master branch. Built a reproducible one-script workflow that automates the PETSc/PFLOTRAN solver-stack compilation process for specialists across the team.
- **Improved the performance of PFLOTRAN's Nernst–Planck-based electromigration implementation**, reducing runtime from approximately 60 minutes to 10 minutes ($\sim 6\times$ speed-up) in an internal project case; further validation on larger project cases is ongoing.
- **Provide technical leadership and client-facing delivery**: onboarded 3 colleagues in numerical modelling, coordinate regular PFLOTRAN technical meetings, and delivered a 3-day technical training course in Tokyo for Obayashi Corporation. Conceived and coordinated a predictive-modelling R&D proposal involving Amphos21, UPC and RSK, submitted to Shell and currently under evaluation.

Apr 2022 – Jul 2024

Recognized Researcher (R2) / Research Engineer, Barcelona Supercomputing Center (BSC)

- **Developed high-fidelity electrochemical multiphysics models** for solid-oxide fuel cells and electrolyzers, coupling electrochemical, thermal and fluid-flow processes within BSC's Alya framework.
- **Designed and implemented adaptive numerical methods** for large-scale simulations on MareNostrum, contributing scientific software to production HPC workflows and to peer-reviewed research on electrochemical transport.
- **Served as BSC's technical point of contact** for the high-fidelity simulation work package of a European R&D consortium, coordinating technical interaction with project partners.

Jun 2021 – Mar 2022

Software Engineer, Lifeware AG, Barcelona

- **Developed and maintained production software** for life-insurance applications in Smalltalk, applying continuous integration and test-driven development to improve code quality and reliability.

Jan 2016 – May 2021

Postdoctoral & PhD Researcher, Universitat Politècnica de Catalunya (UPC)

- **Postdoctoral researcher (2020–2021):** developed high-order numerical formulations and simulation software for computational modelling of glioblastoma growth.
- **PhD researcher (2016–2020):** developed high-order HDG and implicit Runge–Kutta methods for single- and multiphase porous-media flow.

SELECTED PUBLICATIONS

Six peer-reviewed journal articles and book chapters in total, and 10+ contributions to international conferences (EHEC, ECCOMAS, WCCM, ICIAM). Full list: ORCID · Google Scholar.

Costa-Solé, A. et al. (2026). An adaptive numerical framework for the electro-chemical transport in direct internal reforming solid oxide fuel cells (DIR-SOFC). *International Journal for Numerical Methods in Engineering*. doi.org/10.1002/nme.70310

Costa-Solé, A. et al. (2026). High-performance computing for subsurface flow and transport modelling of Olkiluoto. *Procedia Computer Science*, 286, 48–57. doi.org/10.1016/j.procs.2026.08.018

Costa-Solé, A., Ruiz-Gironés, E., Sarrate, J. (2021). High-order HDG formulation with fully implicit temporal schemes for the simulation of two-phase flow through porous media. *International Journal for Numerical Methods in Engineering*, 122(14), 3583–3612. doi.org/10.1002/nme.6674

EDUCATION

2025

MSc in Artificial Intelligence, Universidad Internacional de La Rioja (UNIR). Advanced training in machine learning, deep learning, data-driven modelling, optimization and artificial intelligence for complex computational problems. **Minor Thesis**: Predicción y diagnóstico del Trastorno Obsesivo- Compulsivo mediante técnicas de Inteligencia Artificial aplicadas al análisis de experiencias vitales estresantes (UNIR - Hospital Universitari de Bellvitge).

2020

PhD in Civil Engineering, Universitat Politècnica de Catalunya (UPC), *Cum laude*. **Thesis**: high-order hybridizable discontinuous Galerkin formulations with fully implicit Runge–Kutta schemes for single- and two-phase flow through porous media.

2015

MSc in Geotechnical Engineering, UPC & **MSc in Mineral Resources & Geological Hazards**, Universitat de Barcelona (UB)–Universitat Autonoma de Barcelona (UAB). Advanced interdisciplinary training in geotechnical engineering, hydrogeology, geomechanics, geochemistry, geological modelling and subsurface processes, with a strong emphasis on numerical modelling.

2013

BSc in Geological Engineering, UPC–UB. Engineering degree combining geotechnics, hydrogeology, numerical modelling, subsurface processes and environmental engineering, with a strong focus on quantitative analysis and computational modelling.

TEACHING, SUPERVISION & LANGUAGES

Adjunct Lecturer, UPC (2022–2023, 2026–present): statistics, probability, numerical methods, MATLAB and R programming.

Co-supervisor of an MSc thesis (BSC–UPC, 2024): high-order continuous Galerkin formulation for glioblastoma growth simulation. hdl.handle.net/2117/409235

UPC internal research grants: geostatistical study of the Salar de Atacama with direct involvement with the client (SQM); teaching support.

Ongoing collaborations: development of a simulation code for metamaterials with two UPC faculty members, and an extension of the MSc thesis work with a psychiatrist at Hospital Universitari de Bellvitge.

Catalan, **Spanish**: native. · **English**: advanced professional proficiency.

CERTIFICATIONS & TRAINING

NVIDIA Deep Learning Institute, Fundamentals of Accelerated Computing with CUDA Python In progress, 2026

University of Colorado System / Coursera, Kalman Filter Boot Camp (and State Estimation) In progress, 2026